

Estimation of Quantum Fisher Information by variational quantum circuits on a quantum computer

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In collaboration with the Italian Space Agency (ASI)

Project: *Quantum Computers for Space Exploration*



Applications

Study QFI and TQFI for two applications:

Quantum magnetometry

Estimate a magnetic field with an interacting spin system.

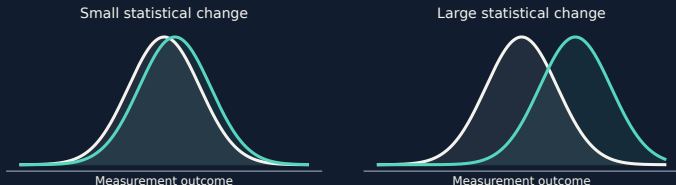
Exoplanet detection

Resolve a small exoplanet with spatial-mode measurements.

A common workflow combining numerical simulation, quantum circuits and experimental data.

Classical Fisher Information

Local sensitivity, distinguishability of nearby parameter values.



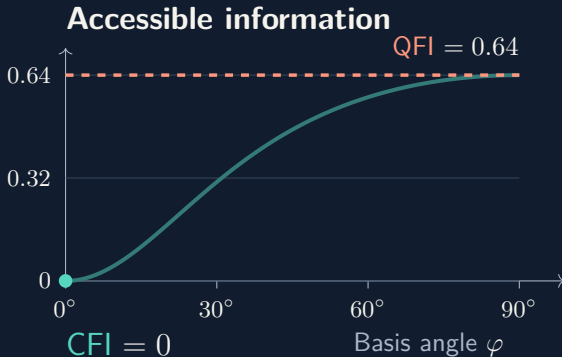
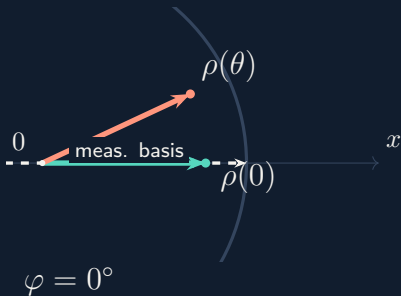
Fisher Information

Sensitivity of an observed probability distribution to changes in an unknown parameter

How much information **a specific, fixed measurement** reveals about an unknown parameter governing a probability distribution.

Fisher Information and Measurements

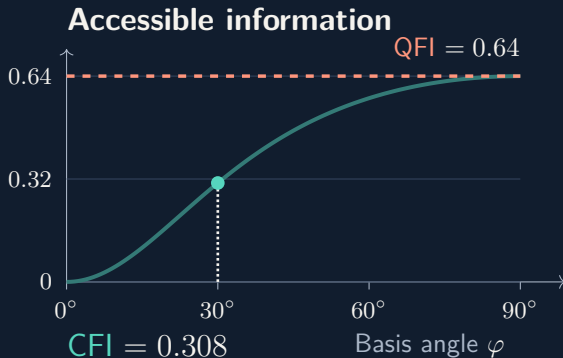
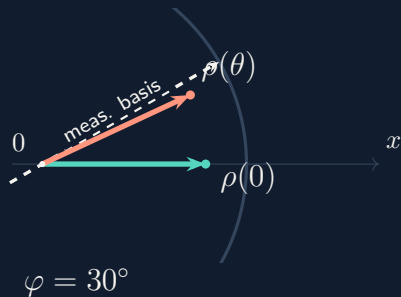
Two states inside the Bloch sphere



Mixed states, $r = 0.8$; finite separation shown for clarity. CFI is local at $\theta = 0$ (radians), not a two-state finite difference. Appendix J.

Fisher Information and Measurements

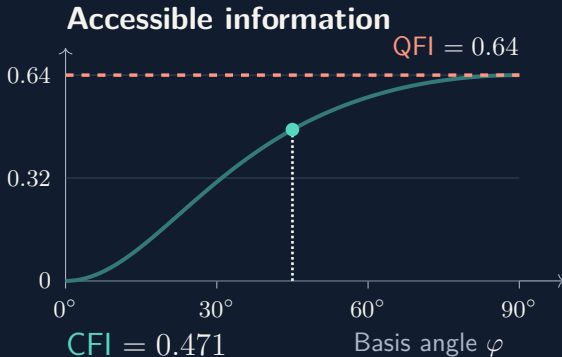
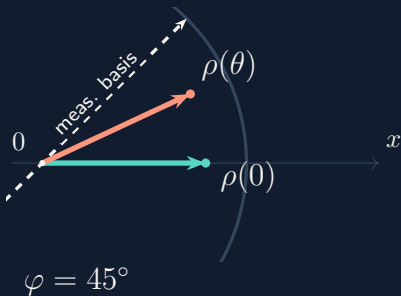
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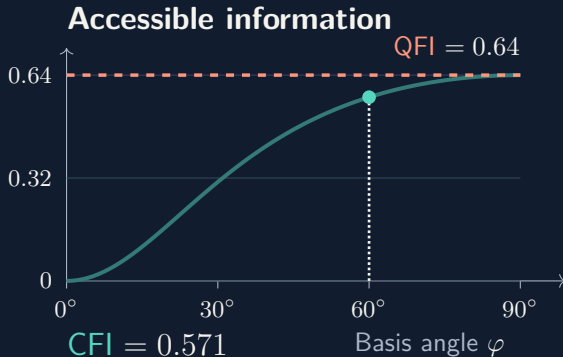
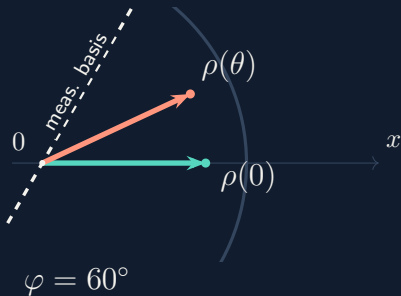
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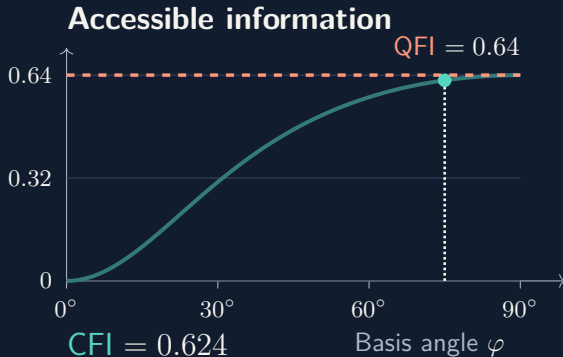
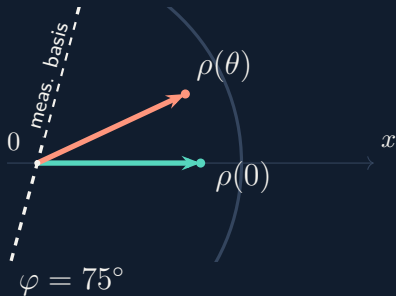
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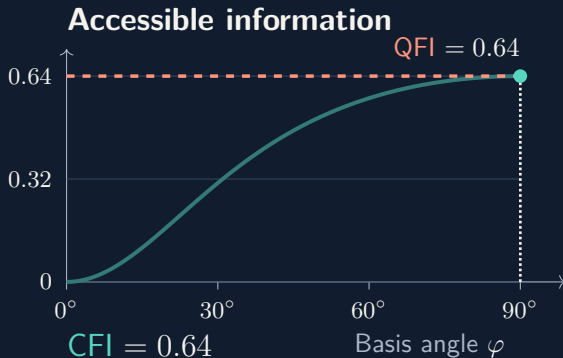
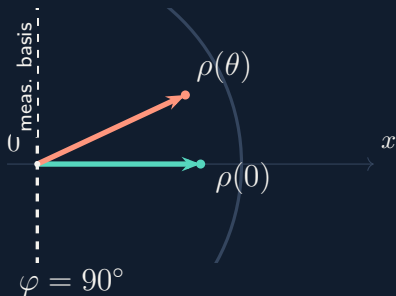
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Fisher Information and Measurements

Two states inside the Bloch sphere



QFI is intrinsic to the quantum state
and it is **independent of the measurement basis**

Mixed states, $r = 0.8$; finite separation shown for clarity. CFI is local at $\theta = 0$ (radians), not a two-state finite difference. Appendix J.

How we treat mixed states

WHAT Statistical mixtures of quantum states, described by a density matrix ρ

Thesis: Introduction. Cerezo et al. (2022); Beckey et al. (2022). Subnormalized truncation; finite-step scope: Appendix A.

How we treat mixed states

WHAT Statistical mixtures of quantum states, described by a density matrix ρ

WHY Real sensors are noisy and only partly accessible
Mixed-state QFI remains less explored and costly to compute exactly

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How we treat mixed states

WHAT Statistical mixtures of quantum states, described by a density matrix ρ

WHY Real sensors are noisy and only partly accessible
Mixed-state QFI remains less explored and costly to compute exactly

HOW Use truncated spectral decomposition

$$\rho_m = \sum_{i=1}^m \lambda_i |\psi_i\rangle \langle \psi_i|$$

The m largest eigenvalues
and their eigenvectors.

VQSE — Variational Quantum State Eigensolver

Learns the leading eigenvalues and a circuit that prepares their eigenvectors.

TQFI — Truncated Quantum Fisher Information

Uses the retained components to construct fidelity-based QFI bounds.

Thesis: Introduction. Cerezo et al. (2022); Beckey et al. (2022). Subnormalized truncation; finite-step scope: Appendix A.

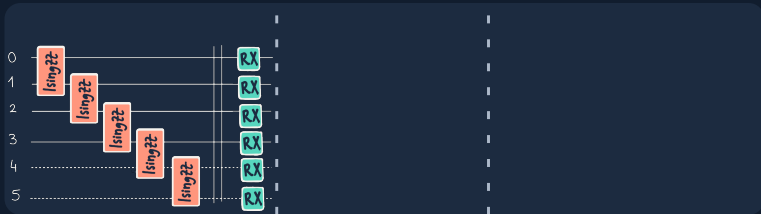
Parallel numerical and quantum-circuit workflow

NUMERICAL SIMULATION

Preparation

Hamiltonian and
time evolution

QUANTUM CIRCUIT



Schematic circuit layouts and example output. Implementations can be exchanged at matched interfaces; physical stage order is preserved.

Parallel numerical and quantum-circuit workflow

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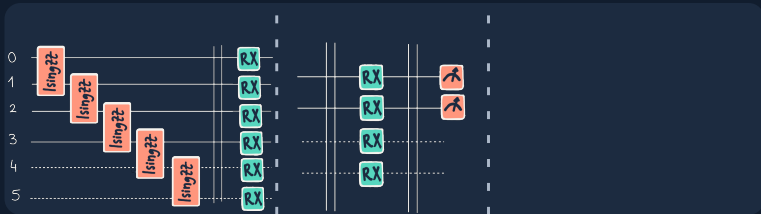
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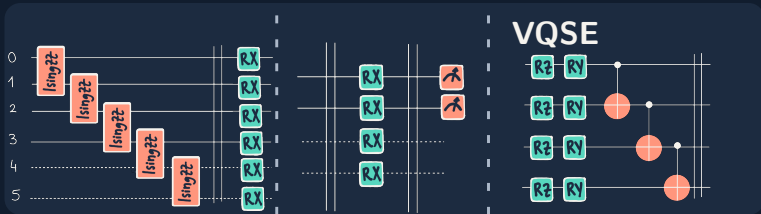
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Exact spectral
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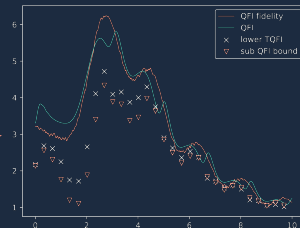
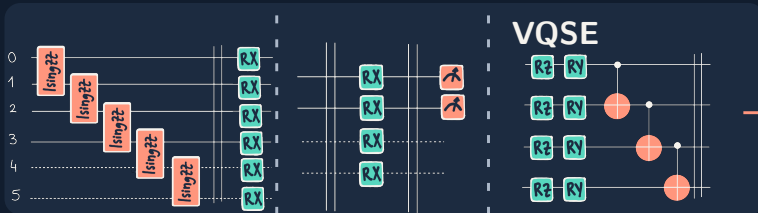
Spectrum

Exact spectral
decomposition

Information

- 1) Exact SLD QFI
- 2) TQFI bounds

QUANTUM CIRCUIT



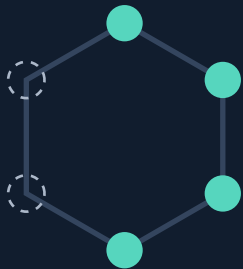
Schematic circuit layouts and example output. Implementations can be exchanged at matched interfaces; physical stage order is preserved.

APPLICATION I

Quantum Magnetometry

A noisy quantum magnetometer

An interacting-spin model for comparing preparation, noise and restricted access.



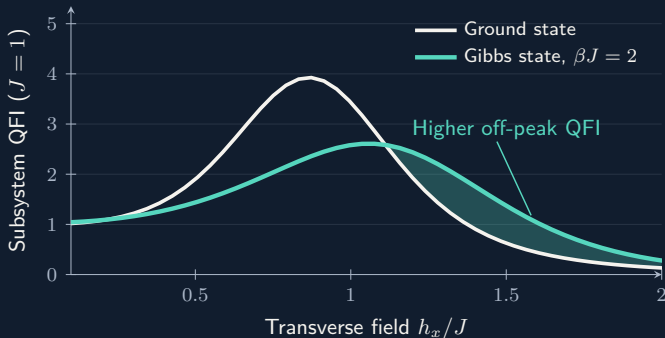
Periodic spin chain
Partial Trace: 2 discarded

Transverse-field Ising model (TFIM)

Ground state	Lowest-energy preparation
Thermal	Field-dependent populations
Depolarized	Fixed depolarizing noise

$N = 6$, $n = 4$, periodic boundaries, $J = 1$. Thesis: NoiseAnalysis.

Thermal preparation broadens the sensitive range



Broader sensitivity

More QFI beyond the ground-state peak.

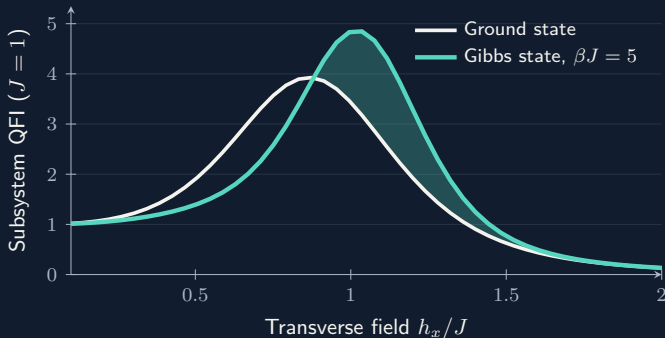
Uncertain field

Useful across a wider field range.

At $\beta = 2$, thermal preparation retains more information beyond the ground-state peak.

Exact SLD QFI; periodic $N = 6$, $n = 4$, $J = 1$. Different preparation families. Thesis: NoiseAnalysis.

Cooling strengthens the sensitivity peak



Stronger peak

Higher QFI around the thermal optimum.

Known field range

Useful near the target field.

At $\beta = 5$, the thermal state exceeds the ground-state peak in the same spin subsystem.

Exact SLD QFI; periodic $N = 6$, $n = 4$, $J = 1$. Different preparation families. Thesis: NoiseAnalysis.

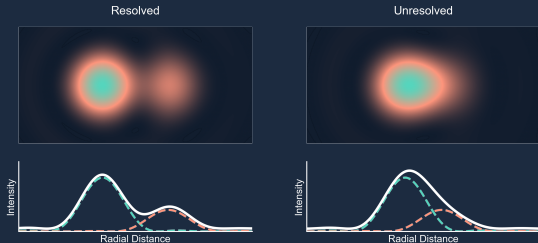
APPLICATION II

Exoplanet Detection

The Exoplanet Problem

A faint exoplanet must be resolved in the presence of a bright star.

Direct imaging, Rayleigh limit



Spatial-mode sorting



Photon counts in a spatial-mode basis.

SPADE changes the measurement basis to reveal weak spatial signals.

SPADE: SPAtial-mode DEMultiplexing. Tsang et al. (2016); Santamaria et al. (2024).

SPADE modes and readout cross-talk

p_0 : fundamental mode

Probability of a photon in HG_{00} dominated by the aligned star

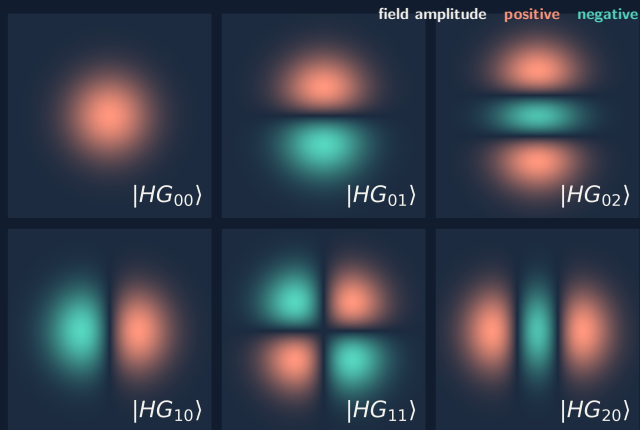
p_1 : first-order modes

Probability in HG_{01} or HG_{10} sensitive to exoplanet separation

χ : readout cross-talk

Probability of reporting port 1 instead of 0, or vice versa

$$p_1^{\text{rep}} = \chi p_0 + (1 - \chi) p_1$$



Modal outcomes, not photon origin. Symmetric readout model; other outcomes unchanged. Thesis: Exoplanet Experiment.

When modal counts become spectral information

SPADE populations are eigenvalues only under a specific basis assumption.

1. MEASURE

$$p_j = \langle HG_j | \rho | HG_j \rangle$$

SPADE returns populations in the chosen HG basis.



2. ONLY IF

$$\rho = \sum_j \lambda_j |HG_j\rangle \langle HG_j|$$

The state is diagonal in HG, and retained modes are individually resolved and normalized.



3. RECONSTRUCT

$$\rho_r = \sum_{j=0}^{r-1} p_j |HG_j\rangle \langle HG_j|$$

Then $p_j = \lambda_j$ for the retained modes and $\text{rank}(\rho_r) \leq r$.

Without the diagonal-basis assumption, modal populations are not the eigenvalues of the state.

A modal cutoff and a spectral rank truncation coincide only when the HG basis diagonalizes the retained state.

From two incoherent sources to an optical state

The state is defined before loss, measurement and detector processing.

PHYSICAL SOURCE

- **centred star**
 $|HG_{00}\rangle$
- **displaced companion**
 $|\psi_{d_a}\rangle$

ONE-PHOTON SOURCE STATE

$$\rho_S(d_a, \epsilon) = (1 - \epsilon)|HG_{00}\rangle\langle HG_{00}| + \epsilon|\psi_{d_a}\rangle\langle\psi_{d_a}|$$

$$|\psi_{d_a}\rangle = \sum_n c_n(d_a)|HG_n\rangle$$

No interference term between the two sources; the displaced component can still contain coherence between HG modes.

The quantum state contains modal structure that the recorded probability p_1 alone cannot reconstruct.

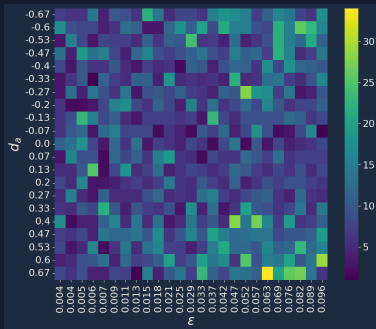
ϵ : faint-source photon fraction. Gaussian PSF and bright-source alignment are source-model assumptions. Thesis: Astrophysical and Exoplanet Experiment.

Experimental data

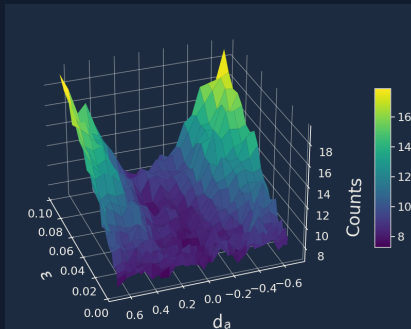
Aggregate first-order counts over separation and source intensity.

52,500 count observations: 525 settings $\times$ 100 repetitions; 10 ms each

ONE REPETITION



MEAN OF 100



Experimental setup

- ▶ d : signed source separation
- ▶ ϵ : relative source intensity
- ▶ Each pixel: aggregate counts assigned to $|HG_{01}\rangle$ and $|HG_{10}\rangle$

Why use quantum channels?

Optical effects act on the state; readout cross-talk acts on probabilities.

$$\rho_S \xrightarrow{\mathcal{E}} \rho' \xrightarrow{\text{SPADE}} \mathbf{p} \xrightarrow{R_\chi} \mathbf{p}^{\text{rep}} \xrightarrow{\text{calibration}} \mu$$

QUANTUM CHANNEL

$$\begin{pmatrix} q_a & c \\ c^* & q_b \end{pmatrix} \xrightarrow{\mathcal{D}_v} \begin{pmatrix} q_a & vc \\ vc^* & q_b \end{pmatrix}$$

Before SPADE: dephasing scales c by v while leaving two HG-mode populations unchanged.

Other optical maps describe loss, alignment, coherent mixing and modal transmission.

CLASSICAL CROSS-TALK

$$\mathbf{p}^{\text{rep}} = R_\chi \mathbf{p}$$

$$p_1^{\text{rep}} = \chi p_0 + (1 - \chi) p_1$$

After SPADE: cross-talk redistributes outcomes after coherence is inaccessible to this measurement.

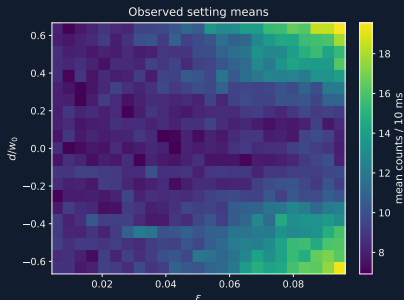
A probability-only model cannot describe pre-measurement coherence or its loss.

The optical channels are specified physical hypotheses, not reconstructed from the aggregate counts; the observable count response is calibrated.

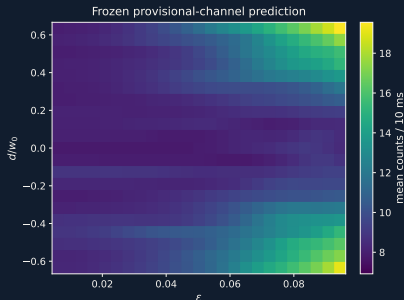
A conditional model captures the observed mean pattern

Both maps show a central minimum and larger counts at greater separation and source ratio.

Observed setting means



Modelled setting means



Agreement supports the mean-response model, not a unique reconstruction of the optical channels.

Thesis appendix: provisional channel workbench, top panels. In-sample mean comparison; optical parameters are conditional assumptions.

Conclusions

Simulation, preparation studies, data interpretation and a variational optical proposal.

- **Numerical and quantum-circuit sensing pipeline**

Simulated quantum sensors under noise and decoherence, estimating QFI and TQFI.

- **Preparation and noise models**

Compared ground and Thermal states, depolarization and restricted access to the sensor.

- **Quantum-to-classical model of the SPADE receiver**

Specified optical channels and calibrated the observable mean-count response.

- **VQ-SPADE: a proposed optical eigensolver**

A trainable mode rotation and SPADE readout would diagonalize the optical state.

Thesis: Methods, Noise Analysis, Exoplanet Experiment and Conclusions. VQ-SPADE is proposed for a future experiment.

Spectral truncation and the local-QFI limit

Dominant eigencomponents bound a fidelity quantity at finite parameter displacement.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

F is root fidelity.

The retained eigenspace is state dependent; it is not a fixed detector-mode cutoff.

At regular operating points, $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$ as $\Delta \rightarrow 0$.

Finite-step bounds need not be pointwise bounds on the exact SLD QFI.

Thesis: Methods; Mathematical Background. Cerezo et al. (2022); Beckey et al. (2022).

The ansatz metric and its scope

The physical sensing parameter and the trainable circuit controls define different derivatives.

$$F_{\mu\nu}^{\text{ans}} = 4 \operatorname{Re}[\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle]$$

Pure ansatz: $F^{\text{ans}} = 4g_{\text{FS}}$, the Fubini–Study metric up to convention.

$$\min_{\Delta\alpha} \nabla C^\top \Delta\alpha + \frac{1}{2\eta} \Delta\alpha^\top (F^{\text{ans}} + \lambda I) \Delta\alpha$$

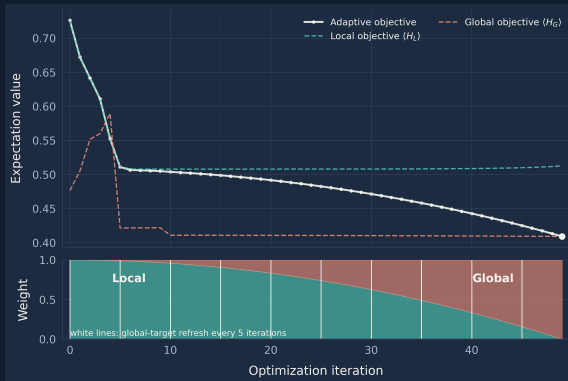
Stationarity gives the regularized natural-gradient update.

The quadratic term constrains state change, not objective curvature.

Mixed VQSE inputs require a mixed-state metric; a pure-state metric is otherwise a surrogate.

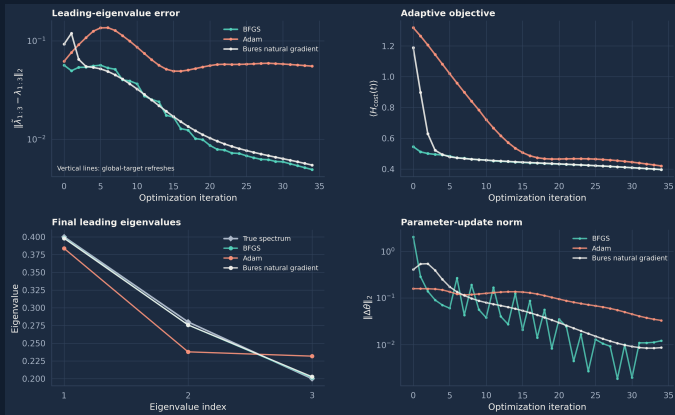
Thesis: Mathematical Background; VQSE. A full metric has quadratic parameter-count overhead; no general training speedup is claimed.

Local-to-global objective schedule



Deterministic two-qubit VQSE example: balanced non-diagonal mixed state, rank-3 target, 50 outer iterations and global-target refresh every 5 iterations.

BFGS, Adam and Bures natural gradient



Deterministic simulator benchmark: non-diagonal two-qubit state with spectrum (0.40, 0.28, 0.20, 0.12), rank-3 target, equal initialization, 35 outer iterations and target refresh every 5 iterations.

Thermal sensitivity and data processing

The Thermal and ground-state families need not have the same field response.

$$\rho_{\beta}(h_x) = \frac{e^{-\beta H(h_x)}}{\text{Tr } e^{-\beta H(h_x)}}$$

The field changes both eigenvectors and thermal populations.

Partial trace still obeys data processing within each family:

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h_x)] \leq \mathcal{F}_Q[\rho_{\beta}(h_x)]$$

For an uncertain field, choose an operating range or prior.

These curves do not establish a universal Bayesian or minimax optimum.

The comparison changes the preparation family; it does not reverse a fixed noise channel.

Thesis: Noise Analysis. Finite system; exact SLD QFI.

Proof of Kraus-unitary freedom

A change of Kraus basis leaves the channel unchanged.

$$\widetilde{K}_a = \sum_b u_{ab} K_b, \quad U^\dagger U = I$$

$$\sum_a u_{ab} u_{ac}^* = \delta_{bc}$$

Orthonormal columns of U .

$$\sum_a \widetilde{K}_a \rho \widetilde{K}_a^\dagger = \sum_{b,c} \delta_{bc} K_b \rho K_c^\dagger = \sum_b K_b \rho K_b^\dagger$$

Isometry in the Kraus index leaves the action on every input unchanged.

A physical realization adds assumptions about the environment; the reduced map alone does not select them.

Standard Kraus equivalence: an isometry, or a unitary after padding with zero operators. Algebra shown explicitly for this presentation.

A classical response does not fix a quantum channel

Embedding a stochastic map is a model construction, not process tomography.

$$\mathbf{p} \mapsto \text{diag}(\mathbf{p}), \quad \mathbf{p}' = R\mathbf{p}$$

For a column-stochastic R , one valid quantum extension is

$$K_{ji} = \sqrt{R_{ji}}|j\rangle\langle i|, \quad \mathcal{E}_R(\rho) = \sum_{ji} R_{ji} \rho_{ii} |j\rangle\langle j|$$

This extension measures and prepares in the chosen basis, erasing coherences.

For $R = I$, both the identity channel and complete dephasing reproduce the same populations.

Different channels can agree on the measured populations; this is distinct from Kraus freedom of one channel.

Thesis: classical/quantum embedding and HG coherence discussion. Channel-extension example derived for the presentation.

The loss map and its physical checks

The input mode space is embedded in an output space containing an orthogonal failure flag.

$$\mathcal{L}_\tau(\rho) = K_t \rho K_t^\dagger + \text{Tr}[(I - K_t^\dagger K_t)\rho] |\text{fail}\rangle \langle \text{fail}|$$

For any ancillary system, Kraus form preserves positivity:

$$(\mathcal{L} \otimes \text{id})(X) = \sum_a (K_a \otimes I) X (K_a^\dagger \otimes I) \geq 0 \text{ for } X \geq 0.$$

Boundary checks: $\tau_j = 1$ transmits mode j ; $\tau_j = 0$ sends it to failure.

Normalization: transmission plus failure has unit trace.

Physical validation: mode-injection measurements determine the τ_j .

Renormalizing the transmitted branch alone changes the question to postselected performance.

Thesis: mode-dependent loss and flagged mode reduction. The record lacks independent modal-transmission calibration.

Predictive calibration of the observed response

A physical mean with a Gaussian-process correction predicts held-out setting means.

$$\mu(d_a, \epsilon) = \mu_0(d_a, \epsilon) \exp[f(d_a, \epsilon) + \xi]$$

Grouped five-fold validation; all components refitted within each training fold.

Held-out structure	RMSE	Standardized SD	95% coverage
Checkerboard cells	0.606	1.004	93.52%
Distance rows	0.734	1.041	94.67%
Source-ratio columns	0.609	1.009	93.71%

The validated target is the mean of 100 repetitions; this does not identify a complete quantum channel.

Thesis: Exoplanet Experiment Appendix, hierarchical calibration validation. RMSE in counts per 10 ms.

From SPADE to a variational optical measurement

VQ-SPADE is a proposed optical realization of variational state eigensolving.

$$p_k(\boldsymbol{\alpha}; d_a) = \langle HG_k | V(\boldsymbol{\alpha}) \rho(d_a) V^\dagger(\boldsymbol{\alpha}) | HG_k \rangle$$

A programmable coherent mode transformation provides the trainable basis.

Resolved outputs, flux and failure events provide normalized probabilities.

Nearby states and additional overlap measurements support fidelity-based QFI bounds.

The present record motivates the acquisition design; it does not implement this protocol.

Thesis: proposed VQ-SPADE scheme. Additional independent channel and background calibrations are required.

Mixed states and measurement-dependent information

Two finite-separated vectors illustrate one state family; Fisher information is evaluated locally.

$$\rho(\theta) = \frac{I + r(\cos \theta X + \sin \theta Z)}{2}, \quad r = 0.8, \quad \rho \equiv \rho(0)$$

$$M_{\pm}^{(\varphi)} = \frac{I \pm (\cos \varphi X + \sin \varphi Z)}{2}, \quad p_{\pm}(\theta) = \frac{1 \pm r \cos(\theta - \varphi)}{2}$$

$$\mathcal{F}_C(\varphi)|_{\theta=0} = \frac{r^2 \sin^2 \varphi}{1 - r^2 \cos^2 \varphi} \leq r^2 = \mathcal{F}_Q$$

At $\varphi = 0^\circ$: CFI = 0. At 45° : CFI = $8/17 \simeq 0.471$.

At $\varphi = 90^\circ$, measuring Z attains QFI = 0.64; the SLD is $L_0 = rZ$.

Both displayed vectors have radius 0.8; their 25° separation is only illustrative.

Changing the measurement changes accessible information, not the underlying quantum limit.

Exact mixed-qubit model; θ in radians. Full rank keeps every displayed measurement regular. No finite-difference QFI approximation is used.

Kraus representations are not unique

Different operator sets can describe exactly the same quantum channel.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \quad \sum_a K_a^\dagger K_a = I$$

Example: complete dephasing of a two-mode state in the chosen basis.

Basis projectors

$$K_0 = |0\rangle\langle 0|, \quad K_1 = |1\rangle\langle 1|$$

Random phase flips

$$\widetilde{K}_0 = \frac{I}{\sqrt{2}}, \quad \widetilde{K}_1 = \frac{Z}{\sqrt{2}}$$

$$\sum_a K_a \rho K_a^\dagger = \frac{\rho + Z\rho Z}{2} = \text{diag}(\rho)$$

Kraus freedom changes the representation. Physical hypotheses concern the channel.

Two-mode illustration of the thesis HG-dephasing model at $v = 0$. Equivalent representations cannot be distinguished by system-only data.

A physical construction: mode-dependent loss

Mode transmission $0 \leq \tau_j \leq 1$; loss is represented by an orthogonal failure state.

Choose operators with explicit physical roles

$$K_t = \sum_j \sqrt{\tau_j} |j\rangle \langle j|, \quad K_j^f = \sqrt{1 - \tau_j} |\text{fail}\rangle \langle j|$$

Transmission and failure jointly account for every input photon.

$$K_t^\dagger K_t + \sum_j (K_j^f)^\dagger K_j^f = \sum_j [\tau_j + (1 - \tau_j)] |j\rangle \langle j| = I$$

Kraus form ensures complete positivity; completeness proves trace preservation.

Channel validity can be proved mathematically; physical suitability requires calibration.

Derived from the thesis mode-dependent loss map. Input: one-photon mode space after common survival; failure is retained in the output.

Information retained by the optical receiver

Compare quantum and classical information at successive stages of the receiver.

$$\mathcal{F}_Q(\rho_S) \geq \mathcal{F}_Q[\mathcal{E}(\rho_S)] \geq \mathcal{F}_C(\mathbf{p}) \geq \mathcal{F}_C(R\mathbf{p})$$

For channels and measurements independent of the parameter being estimated.

Checkpoint

Fraction of source QFI

● Resolved reference, first-order and failure outputs	99.68–100%
● Binary readout; assumed 0.35% port confusion	0.13–57.0%

Aggregation and readout confusion can strongly reduce the accessible information.

Conditional model: same nine settings, per photon surviving common loss. Ranges across settings; not reconstructed experimental QFI.

From information to precision: the quantum Cramér–Rao bound

QFI is the best local Fisher information available over all quantum measurements.

FIXED POVM

$$\text{Var}(\hat{\theta}) \geq \frac{1}{N\mathcal{F}_C(\theta; M)}$$

Assuming N independent trials and local unbiasedness.

BEST MEASUREMENT

$$\mathcal{F}_C(\theta; M) \leq \mathcal{F}_Q(\theta)$$

Maximize the CFI over all POVMs for the same state family.

QUANTUM LIMIT

$$\text{Var}(\hat{\theta}) \geq \frac{1}{N\mathcal{F}_Q(\theta)}$$

Equivalently, $\Delta\theta \geq 1/\sqrt{N\mathcal{F}_Q}$.

The QCRB is an ideal local precision limit; attaining it also requires a suitable measurement and estimator.

Assumptions: differentiable one-parameter model, independent repetitions, local unbiasedness and regularity. The optimal measurement can depend on the unknown parameter. Thesis: Mathematical Background.